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Protocol for a national intervention programme aimed to reduce unwarranted variation and overuse of shoulder arthroscopy in Norway

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ABSTRACT

Several advice-based approaches have aimed to reduce the overuse of procedures characterised as low-value healthcare services but have so far shown marginal impact on utilisation rates. Differences in utilisation rates tend to be associated with type of healthcare provider. Commercial providers have a tendency to have higher utilisations rates compared with non-profit providers. Norway has a publicly funded universal health coverage system where healthcare services are mainly provided by publicly owned hospitals in addition to a small number of non-profit hospitals and commercial hospitals reimbursed by the regional health authorities. According to previous experiences, unwarranted variation in utilisation rates can be associated with the type of healthcare provider. The aim of the present study is to evaluate the impact of tailoring the interventions on reducing overuse of low-value care based on an analysis of the clinical pathways and the various types of healthcare providers. The National Decision Forum in Norway has decided that acromion resection in patients with impingement symptoms and repair of non-traumatic cuff rupture should not be offered as standard treatment. The present protocol describes the planned intervention to reduce the overuse of these procedures.

INTRODUCTION

Reducing utilisation rates for low-value care is a key element in improvement work that benefits patients and reduces wasteful, unnecessary or even harmful care.[1] During the last decades, several advice-based approaches such as 'Choosing Wisely' and the 'Evidence-Based Intervention programme (EBI)' have aimed to reduce the overuse of procedures characterised as low-value healthcare service but have so far shown marginal results on utilisation rates.[2 3]

Utilisation rates for healthcare services vary both within and between nations. Although data from different national health atlases for methodical reasons may be difficult to compare, they tend to reveal differences in utilisation rates associated with the type of healthcare system; commercial providers have a tendency to have higher utilisation rates compared with public non-for-profit providers. However, even in countries with publicly funded universal healthcare such as Norway, utilisation rates for medical and surgical interventions vary between and within health regions and hospital areas, which may be caused by the type of healthcare provider.

Interventions targeting overuse and unwanted variation should be based on an understanding of which part of the variation is unwanted. So far, most interventions targeting overuse are based on the assumption that knowledge and data revealing unwarranted variation eventually will lead to a reduction in unwarranted variation, assuming an effect independent of the type of healthcare provider. If utilisation rates for low-value healthcare reflect variations that appear to be influenced by the type of healthcare provider, this may also be considered as unwanted variation.

Norway has a publicly funded universal health coverage system where healthcare services are mainly provided by publicly owned hospitals in addition to a small number of non-profit hospitals and commercial hospitals reimbursed by the regional health authorities (RHAs).

In 2023, the chief executive officers (CEOs) in the four Norwegian Regional Health Authorites (RHA) launched a programme governed by the Centre for Clinical Evaluation and Documentation (SKDE) to reduce low-value healthcare (The NOR programme). The goal was to reduce unnecessary examinations and treatment within and between the regions. In addition to reducing the use

of upper endoscopy for patients <45 years[4] and the use of coronary angiograms in patients with stable coronary artery disease, arthroscopic shoulder surgery in patients with impingement syndrome and repair of non-traumatic cuff ruptures were selected as the first low-value procedures in the programme.

Based on health technology assessment (HTA) of subacromial decompression surgery[5] and rotator cuff repair[6] published by the Norwegian Institute of Public Health (Folkehelseinistituttet, FHI), Beslutningsforum (The National 'Decision Forum' in Norway) recommended limited use of these interventions and requested the regional chief medical officers (CMOs) to initiate work to limit the use of these procedures. Using experience from previous work on reducing the volumes of knee arthroscopy in South-Eastern Norway Health Authority (HSO),[7] where part of the observed unwarranted variation in utilisation rates could be attributed to the nature of the healthcare provider, we want to explore the effect of tailoring the interventions on reducing overuse of low-value care on appearing variation in the clinical patients and the various types of healthcare providers for patients undergoing these procedures.

It has previously been recommended that poorly documented and potentially harmful surgical practices should be reduced through both professional guidance and administrative regulations.[7] In addition to an analysis of the providers of publicly financed arthroscopic shoulder surgery, we have used data evaluating the clinical pathways to design the intervention. The aim of the planned intervention is to evaluate the effectiveness of a multifaceted set of interventions to reduce the overuse of acromial resection for impingement and suture of non-traumatic cuff repair in the shoulder in a public health system.

Overview of the Norwegian healthcare system and its impact on surgical volumes

The specialist healthcare system in Norway is predominantly a publicly funded universal health coverage system designed to care for about 5.5 million citizens and is provided by four RHA, each responsible for delivering specialist healthcare in their region. The local health trusts in the regions (trusts and private non-profit hospitals) are legal entities governed by independent boards with an overall responsibility for the clinical services they provide.

Most publicly funded shoulder arthroscopies are performed in public hospital trusts and non-profit hospitals in the Western and South-Eastern regions, whereas some procedures are additionally conducted in a few commercial hospitals reimbursed by the RHA in the other regions (Central and North).

Decisions regarding the implementation of new methods and deimplementation of medication and procedural treatments considered as 'low value' are made by 'Nye Metoder' (New Methods), a national system governed by the four RHAs in Norway for decisions regarding implementation of new treatments. 'Beslutningsforum' (The National Decision Forum) comprises the CEOs from the RHAs and makes decisions on the introduction of new treatment and deimplementation of low-value treatment. The Decision Forum prioritises based on the criteria of severity, utility and cost-effectiveness.[8]

Shoulder arthroscopy

The introduction of shoulder arthroscopy has been regarded as one of the greatest improvements in orthopaedic surgery.[9] It was introduced in 1965 for treating adhesive capsulitis.[10] Two decades later, the indications were successively expanded to rotator cuff debridement[11] and subacromial decompression,[12] followed by treatment for shoulder instability,[13] fracture of the humerus and glenoid and rotator cuff repair.[14] The introductions of these new indications were mainly based on data from publications categorised as level 4 evidence, where a majority were reporting case and descriptive series conducted in the USA.[15] Several randomised controlled trials have failed to demonstrate significant beneficial effects of surgical intervention on pain and function compared with exercise interventions for patients with subacromial pain.[16 17] Despite these findings, the volume of acromioplasties in the USA increased by 254%, compared with a 78% increase in the volume of all orthopaedic ambulatory surgery procedures from 1996 to 2006.[18] The same tendency was found in Finland from 1998 to 2007; the rate of subacromial decompression started to decrease at non-profit public hospitals and was down 23% at the end of the study period in 2011 but continued to increase in profit-based private orthopaedic clinics.[19] In National Health Service Scotland, a similar decrease in the volumes of subacromial decompression at public health institutions was observed from 2014 to 2018 by 29%.[20]

Based on a systematic review, and in particular, two sham surgical trials,[21 22] Vandvik *et al*[23] recommended in the BMJ 'rapid recommendation' series in 2019 that patients with atraumatic shoulder pain lasting more than 3 months diagnosed as subacromial pain syndrome, also labelled as rotator cuff disease, '*should avoid subacromial decompression surgery because there is no benefit but there are harms, and it is burdensome*'. The South-Eastern Norway Regional Authority established subsequently an expert group in 2020 which gave the following advice; Patients with non-traumatic shoulder paint should be guided as follows by their general practitioner (GP)

- Surgical treatment is usually not indicated.
- Referral to MRI and orthopaedic surgeon is usually not indicated in the early stage.
- The primary care physician should be aware of this knowledge and recommendation and inform patients accordingly and refer patients for physiotherapy and eventually a specialist in physical medicine and rehabilitation.

The following year, the Norwegian Institute of Public Health (FHI) published a rapid HTA with the following conclusion: There was no clinically relevant difference

between subacromial decompression surgery and conservative treatment for pain, function and health-related quality of life at 1-year follow-up.[5] The same year, the National Decision Forum decided that acromion resection should not be offered as a standard treatment for impingement syndrome.[24]

The same regional expert group questioned the effectiveness of cuff repair in degenerative tears and asked for a national HTA. Responding to their request, FHI published in 2023 a HTA of rotator cuff repair compared with non-surgical treatment for degenerative full-thickness rotator cuff tears published with the following conclusion:[6] For patients with tears involving only one tendon, repair makes little or no clinically relevant difference in pain, night pain, function, patient satisfaction and health-related quality of life compared with non-surgical treatment at 1 year. For patients with tears involving one or two tendons, repair gives slightly higher patient satisfaction, but little or no clinically relevant difference in pain, night pain, function and health-related quality of life compared with non-surgical treatment at 1 year.

The same year, the following decision was made by The National Decision Forum based on the recommendation published by FHI for cuff repair: Conservative treatment of degenerative rotator cuff rupture should be the first choice and sufficiently tested before any surgical intervention.[25] This recommendation was mainly based on two randomised controlled trials[26 27] comparing surgery with non-operative treatment because most randomised trials have compared different surgical methods.[28]

Despite the new clinical guidelines, which were expected to contribute to a reduction in the volumes of shoulder arthroscopies, the utilisation rates remained relatively stable. According to data from the Norwegian Patient Registry (NPR), the utilisation rate for acromion resection in Norway decreased 23% from 69 to 53 per 100 000 inhabitants over the four consecutive years (2019–2022) after the recommendations were published. However, a substantial volume (n=3340) of the procedure was still conducted in 2023. The same year, almost 4400 cuff -repair procedures were conducted, and the utilisation rate was only reduced by 4% in the last 4 years from 77 to 74 procedures per 100 000 a decade after the results from the first randomised trials were published. Furthermore, a significant variation in utilisation rates within and between the regional health regions was observed. Although the national volume of shoulder acromial resections has decreased after 2020 the volume of other arthroscopic shoulder procedures has remained relatively stable during the same period (figures 1 and 2). It is debated whether 'creative coding' or an increase in other surgeries, including procedures whose cost-effectiveness is sparsely documented, partly can explain this trend.

Experience from reducing overuse of knee arthroscopy in Norway

In the last decades, several studies have documented that the effects of knee arthroscopy in patients with degenerative disease are not superior to non-operative treatment and that outcomes after arthroscopic partial meniscectomy in patients with degenerative disease were no better than sham surgical procedures.[29] Siemieniuk *et al*[30] published 'strong recommendation against the use of arthroscopy in nearly all patients with degenerative knee disease' as a part of the BMJ 'rapid recommendation' series. In 2018, the experience with reducing the volumes of knee arthroscopy in South-Eastern Norway Health Authority (HSO) was published, reporting that the observed decrease was caused by both changes in clinical practice following the publication of the first studies showing non-superiority for surgical treatment but also as a result of interventions targeting the healthcare providers.[7] After the South-Eastern Region terminated the contract with a private commercial hospital performing a substantial volume of knee arthroscopy in patients above 50 years, the utilisation rate in the region was reduced by 48% from 2012 to 2016 vs 14% in the other regions which chose to prolong their contracts with the commercial providers offering publicly reimbursed

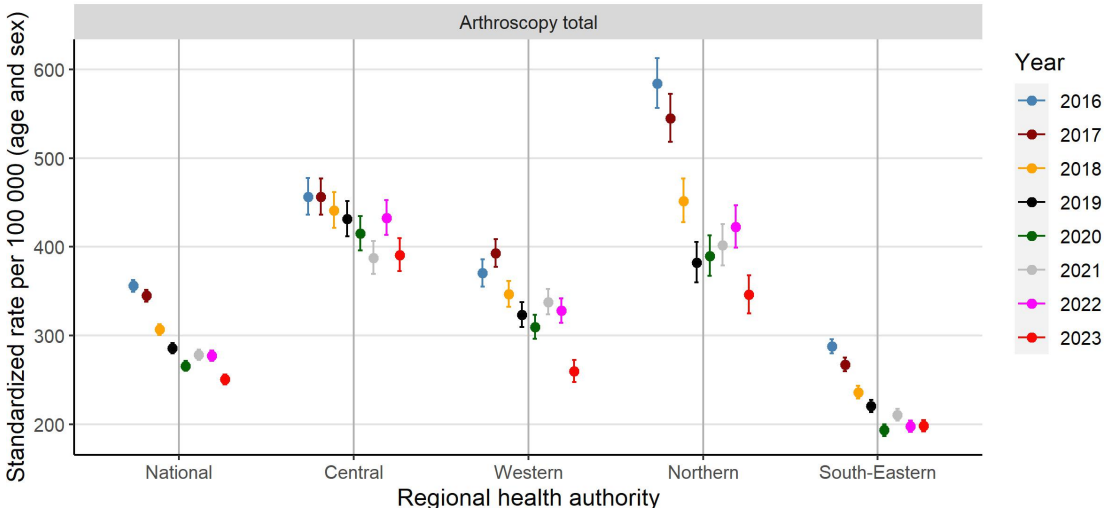

Figure 1 Standardised rates for total arthroscopic shoulder procedures in Norway 2016–2023.

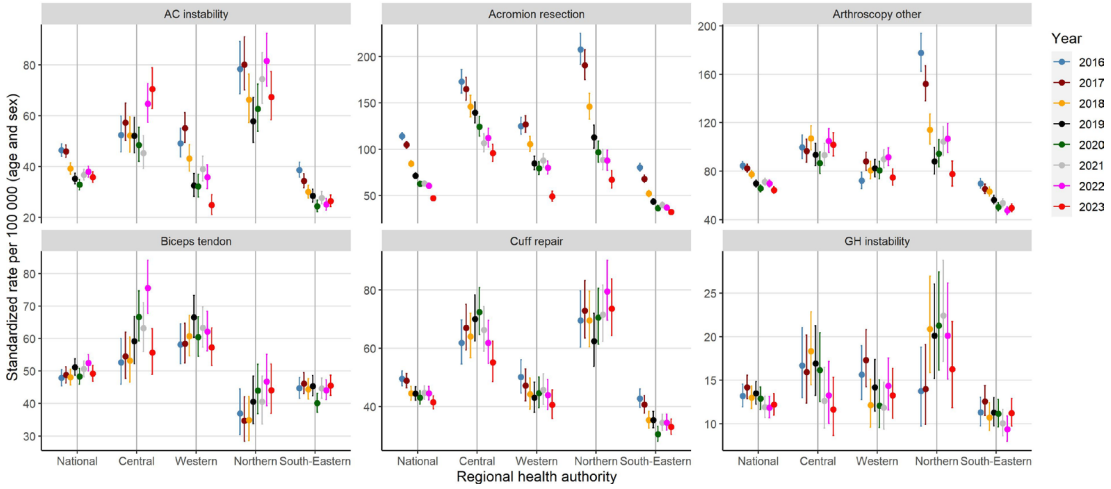

Figure 2 Standardised rates for arthroscopic shoulder procedures in Norway 2016–2023.

Table 1 Age-adjusted utilisation rates per 100 000 inhabitants for acromion resection and cuff repair in Norway (mean and range (hospitals))

	Norway	South-East	West	Central	North
Acromion	68 (7–164)	31 (7–70)	87 (47–132)	121 (58–164)	90 (55–130)
Cuff repair	83 (40–172)	57 (40–82)	99 (48–172)	110 (79–142)	110 (83–144)

knee arthroscopies. Moreover, the introduction of AktivA, a national implementation model of evidence-based guidelines for first-line treatment of patients with hip and knee osteoarthritis, including degenerative meniscus tear provided by trained physical therapists, may have had a major impact on the shift of the clinical pathway from surgical to non-operative treatment.[31]

METHODS AND ANALYSIS

All patients undergoing publicly reimbursed arthroscopic shoulder surgery from 1 January 2025 to 31 December 2026 will be included in the study. The following data sources will be used:

- Data from primary healthcare registries will be used to monitor the number of patients visiting GP, physiotherapist or chiropractor with the International Classification of Primary Care (ICPC) 2 codes L08 Shoulder symptom/complaint and L92 Shoulder Syndrome.
- Data from NPR will be used to monitor following number of patients in public, non-profit and private hospitals with public reimbursement
  - Patients undergoing arthroscopic shoulder procedures.
  - Patients undergoing MRI of the shoulder.
  - Patients referred to the orthopaedic outpatient clinic with non-traumatic shoulder symptoms.

Since the intervention is part of a national project which will be implemented in Norway to reduce utilisation rates of low-value procedures, no sample size has been estimated to evaluate whether the primary outcome is statistically significant or clinically relevant. Utilisation rates will be calculated as direct standardised rates using the R package PHE indicator methods.[32]

Clinical dashboard

The regional South-Eastern Health Trust ('HSO') has developed a dashboard accessible for all hospitals in the region displaying variation in utilisation rates, based on the Nomesco Classification of Surgical Procedures and some radiological procedures that categorise all relevant procedures into a hierarchical system.[33] It has been decided that the Centre for Clinical Evaluation

Table 2 Hospital with the highest volume (percentage of total volume in the region)

	South-East	West	Central	North
Acromion	16% Public hospital	42% Private non-profit	62% Private commercial	42% Private commercial
Cuff repair	22% Private non-profit	42% Private non-profit	43% Private commercial	33% Public hospital

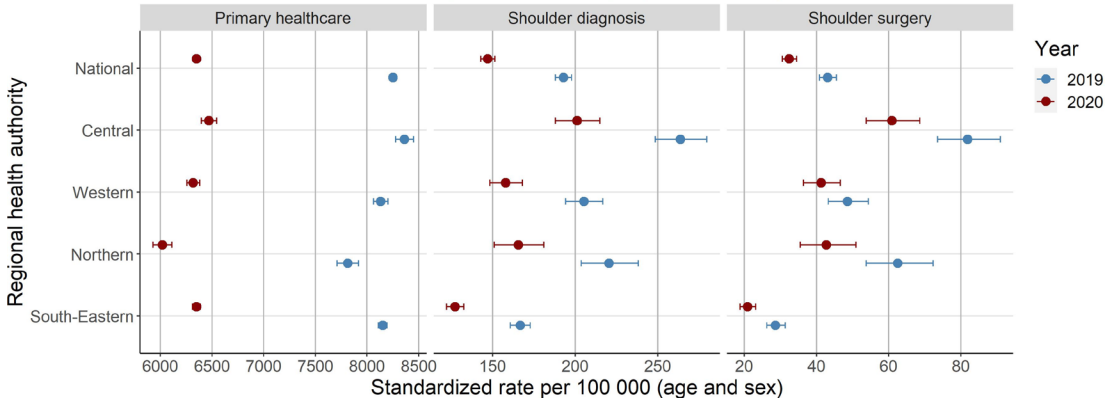

Figure 3 Standardised rates for shoulder complaints in primary healthcare, follow-up in specialised healthcare and surgery with first episode in 2019 and 2020.

and Documentation (SKDE) in Norway will publish the dashboard with national data. The dashboard includes a module that monitors regional unwarranted variation in shoulder arthroscopy with focus on geographical utilisation rates and the specific hospital volumes of the various procedures. The following International Classification of Diseases (ICD) codes are used to identify patients included in the study:

Acromial resection: NBK 13.

Cuff repair: NBL 49.

GH-Instability: NBH 71, NBE 21.

AC instability; NBE31, NBH10, NBH12, NBK12.

Biceps tendon: NBL 39, NBL 69.

Other: other arthroscopic procedures in the shoulder.

Utilisation rates 2023.

Acromial resection for impingement; rate adjusted average rate in Norway; 68 per 100 000 inhabitants with differences between the regions (South East; 31, West; 87, Central; 121, North; 90).

Rotator cuff repair; average rate in Norway; 83 per 100 000 inhabitants with significant differences between the regions (South East; 57, West; 99, Central; 110, North; 110) (table 1).

Healthcare providers

In two of the regional authorities (Central and North), a large portion of the acromial resections were performed in publicly reimbursed private hospitals (62% and 42%, respectively) whereas all procedures were performed in public or non-profit hospitals in the two remaining regional authorities. For cuff repair, the proportion of procedures performed in non-profit hospitals was 22% and 42%, respectively (South-East and West), whereas 43% was done in private commercial publicly reimbursed hospitals (Central), whereas 33% of the procedures were conducted in a public hospital in the Northern region (table 2).

Clinical pathway for patients with shoulder complaints

It has been postulated that the clinical pathway in primary healthcare for patients with non-traumatic shoulder pain might impact the overuse of the defined procedures and that reduction of imaging and the number of referrals to orthopaedic surgeons might have a significant impact on the utilisation rates of these procedures.

However, the patient administrative data we used to monitor the clinical pathway for patients with shoulder complaints in Norway 2019–2023 revealed increased geographical variation on regional level moving from number of patient visits in primary healthcare per 100 000 inhabitants, follow-up at orthopaedic departments, receiving a shoulder diagnosis (ICD-10 M19.x1; M75) or surgery (figure 3).

Intervention

This protocol describes the development and strategy to implement a national intervention aimed at reducing unwarranted variation and overuse of shoulder arthroscopy in Norway. The manuscript describing the protocol is based on the revised Standards for Quality Improvement Reporting Excellence SQUIRE 2.0.

The study is conducted in the Norwegian Health Care system. Although the literature fails to demonstrate the beneficial effect of surgical treatment of impingement and the beneficial effect of degenerative cuff rupture is at best small and debated, we assume that there is a limited number of individuals who might profit from surgical treatment. We have established a multidisciplinary advisory board which will give advice regarding the estimated volume of surgery needed for these patient groups. Furthermore, we plan to explore the value of using available data to assess the association between utilisation rates and average clinical effect of various procedures as described by Rudolfsen *et al*[34] to estimate recommended utilisation rates. Accordingly, the aim will not be to stop this type of surgical treatment but to reduce it to a predefined level which obviously is much lower than it has been until now.

Furthermore, an audit conducted in HSO revealed that the strategic aim of reducing unwanted clinical variation was not clearly communicated by senior HSO management to hospital boards and senior management, and it was recommended that the hospitals achieve a better

understanding of the causes of variation by strengthening their efforts to reduce unwarranted variation in utilisation rates as a key element in improving healthcare quality and patient safety.[35] We have tried to take this advice into account when designing the intervention. The study will be performed in all four RHAs in Norway based on the national recommendations to reduce the overuse of the defined procedures.

In addition to establishing a standardised regime for physiotherapy, we have mainly focused our efforts to reduce the surgical volumes by targeting the surgical activity. Based on the differences between the RHAs in the providers of arthroscopic shoulder surgery, three different interventions will be implemented. The aim of the study is to evaluate three different interventions to reduce the number of arthroscopic acromial resection for patients with impingement and cuff repair for patients with degenerative cuff rupture by various means to harmonise the indications for surgery.

National intervention

Standardisation of non-operative treatment for shoulder disease based on experience from the AktivA model

The AktivA model[31] was initiated in Norway in 2015 based on models in Sweden (BOA—https://boa.registercentrum.se/in-english/the-swedish-osteoarthritis-registry/p/By_o8GxVg) and Denmark (GLAD—https://gladinternational.org/). AktivA is a national model of implementation of evidence-based guidelines for first-line treatment of patients with hip and knee osteoarthritis, including degenerative meniscus tear. All patients go through first-line treatment in the primary healthcare before referral to specialist healthcare. The overall aim of AktivA is to reach all patients in Norway with knee and/or hip osteoarthritis and offer evidence-based first-line treatment regardless of where they live. The AktivA model consists of three parts; an education course for physiotherapists, a patient education programme (an osteoarthritis school) and a supervised, individually tailored exercise programme of 6–12 weeks and the AktivA Quality Registry. A similar programme for patients with non-traumatic shoulder pain will be considered.

Updated guidelines for patients with non-traumatic shoulder pain

The recommendations for patients with non-traumatic shoulder pain from the HSO advisory board will be distributed to the GPs

- Surgical treatment is usually not indicated.
- Referral to MRI and orthopaedic surgeon is usually not indicated at this early stage.

The primary care physician should be aware of this knowledge and recommendation and inform patients accordingly and refer patients for physiotherapy and eventually a specialist in physical medicine and rehabilitation.

Regional interventions

The interventions are designed based on the different patterns of delivery of arthroscopic shoulder surgery in the region health authorities:

Intervention 1: South-Eastern RHA

The utilisation rates in the South-Eastern region are low compared with the national average and the other regions. The surgery is performed in 13 hospitals. As previously described, an advisory board will be asked to estimate the number of procedures reflecting the need for treatment on individual indications. Based on this estimate, it will be considered to appoint a limited number of hospitals which will be asked to establish a multidisciplinary shoulder team with competence for diagnosis and non-operative treatment and evidence-based surgical treatment of shoulder disease causing pain, reduced function, patient satisfaction and health-related quality of life. These hospitals will cover the regional need for this type of surgery which will be reimbursed. The other hospitals will not receive reimbursement for acromial resection and cuff repair.

Intervention 2: Western Norway RHA

Almost half of the surgical volume in the Western region is conducted in one hospital. This hospital will be subjected to clinical audits, and the surgical volumes will be monitored by the regional authority and evaluated on the basis of the estimated volumes described in intervention 1.

Intervention 3: Central and Northern RHA

In the two most northern regions, a substantial part of the volumes is conducted by private hospitals reimbursed by the regional authorities. These contracts will be terminated as soon as it is possible contractually.

Limitations

The interventions described in the present protocol will be implemented in the public health system in Norway, where public healthcare is available to all, regardless of income or geography. Approximately one-fifth of the population have additional private healthcare insurance often covered by the employer. The described interventions do not target healthcare in private hospitals covered by insurance. This might limit the transfer value to other healthcare settings.

Climate effect

At the end of the study, the climate effect of the intervention will be evaluated, based on estimations of the reduction of carbon dioxide emission of patient travel and number of procedures.

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